

Calibrate Once, Judge Anywhere: Multicalibration for Unbiased Aggregate LLM-as-Judge Evaluation under Distribution Shift

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Abstract

Large language models are increasingly evaluated by other language models: an *LLM judge* scores responses for correctness, helpfulness, or safety, and the numbers that drive leaderboards, model-selection decisions, and safety sign-off are *aggregates* over those scores: pass rates, win rates, and violation rates. Because judges are imperfect, a growing literature corrects the aggregate for judge error using estimated sensitivity and specificity (a Rogan–Gladden adjustment). We show that this correction, and confidence-elicitation and post-hoc-calibration alternatives, rest on an invariance that fails in the settings where evaluation matters most: when the population being judged shifts (a new model family, a new task mix, a new language, longer responses), the judge’s error rates shift with it, and the corrected aggregate becomes biased and highly unstable. We show that *multicalibration* (calibration conditional on input features rather than on average) is sufficient for unbiased aggregate estimation under any such covariate shift along the calibrated features, yielding a judge that is calibrated once and applied to any target within its support with no per-target re-estimation. On a controlled benchmark with exact ground truth (employment prevalence from the American Community Survey) and on a large LLM-judge study (Claude Sonnet judging the helpfulness of 17,211 OpenAssistant responses across six languages, graded against human labels), a globally corrected judge is biased by 4–53 percentage points and its estimates carry confidence intervals several times wider than multicalibration’s, whereas the multicalibrated estimator is near-unbiased within support and degrades gracefully out of it. The correction works on the bare Yes/No labels practitioners already collect, matching elicited probabilities, so it comes from the calibration design, not from coaxing confidences out of the judge. High judge agreement (AUC) is neither necessary nor sufficient for a trustworthy aggregate; feature-conditional calibration is what makes LLM-based evaluation transport across the distributions that motivate its use.

Introduction

Evaluating language models at scale increasingly means using a language model to do the evaluating. An *LLM judge* reads a model’s output and scores it (correct or not, helpful or not, safe or not),

and these per-response judgments are aggregated into the summary statistics that decide which model ships: pass rates on a benchmark, win rates in a head-to-head, the fraction of responses flagged unsafe, a helpfulness score on a held-out set (Zheng et al. 2023; Yuan et al. 2024; Lambert et al. 2024). LLM judges are attractive because the quantities they estimate (helpfulness, factuality, safety, instruction-following) are expensive or impossible to verify programmatically, so a cheap automated grader is applied across many models, tasks, and deployments.

The quantity of interest in these workflows is almost never the label on any single response; it is an *aggregate*: what fraction of this model’s answers are correct, how often does that model produce unsafe output, which of two systems wins more often. In the language of the quantification literature, these are *prevalence estimation* problems (González et al. 2017; Hopkins and King 2010), estimating a class proportion in a population from an imperfect classifier. The imperfect classifier is the judge.

The standard way to validate a judge is to report its agreement with human labels: accuracy, F_1 , AUC, or Cohen’s κ on a labeled meta-evaluation set (Zheng et al. 2023; Lambert et al. 2024). A judge that clears a bar is then deployed to score new models and new tasks. A more careful recent literature recognizes that raw agreement is not enough for a *number*: because the judge has imperfect sensitivity and specificity, the raw fraction it labels positive is a biased estimate of the true rate, and this bias should be corrected. The correction that has emerged as the recommended practice is a plug-in Rogan–Gladen adjustment (Rogan and Gladen 1978): estimate the judge’s true-positive and false-positive rates on a labeled calibration set, then invert to debias the reported rate, with confidence intervals that propagate calibration uncertainty (Lee et al. 2026; Fiedler 2026).

We argue that this correction solves the wrong problem for the setting in which judges are actually used. Rogan–Gladen and its relatives assume the judge’s *conditional error behavior is invariant* across the calibration population and the target population being evaluated: that a single pair of sensitivity/specificity numbers estimated once transfers to wherever the judge is applied. This is a label-shift-style invariance: it allows the true positive rate $P(Y=1)$ to change but requires the judge’s error rates given the truth to be fixed. The regime where evaluation actually lives is the opposite. A judge is calibrated on one distribution of responses and then applied to a *shifted* one: a newly released model whose outputs differ stylistically from those in the calibration set, a task mix weighted toward harder problems, a different language, systematically longer answers. Under such *covariate shift*, the judge’s error rates are not constants (they depend on the features of what is being judged), and a correction built on fixed error rates is biased. Worse, when a judge is lenient (as we find LLM judges often are), the Rogan–Gladen ratio has a small denominator and its variance explodes, so even where its point estimate looks acceptable it is not usable.

We show that the property that makes an aggregate transport across populations is *multicalibration*: calibration conditional on the input features, not merely on average (Hébert-Johnson et al. 2018). The relevant guarantee is the “universal adaptability” result of Kim et al. (2022): a predictor that is multi-accurate with respect to a class of feature-defined groups yields unbiased target means under *any* reweighting of the population captured by those groups, with no knowledge of the shift and no target-specific estimation. We translate this into a protocol for LLM-judge evaluation: calibrate a judge once, conditioning on observable metadata (model family, task domain, language, response length) and the judge’s own output, and then read off an unbiased aggregate on any target whose features lie within the calibration support. The correction applies to any judge output, from the discrete Yes/No labels that dominate applied practice to elicited confidence scores, because it operates on the relationship between the judge and the truth *conditional on features*, not on the intrinsic quality of the judge’s scores. We do not claim the underlying theorem; our contribution is (i) the diagnosis that the covariate-shift regime, not the label-shift regime the Rogan–Gladen

family assumes, governs LLM-judge evaluation, and (ii) empirical evidence that feature-conditional calibration fixes it, works on bare labels, and that the fix comes from the calibration design rather than from confidence elicitation.

Three findings anchor the paper. First, on a large LLM-judge study (Claude Sonnet judging the helpfulness of 17,211 OpenAssistant responses across six languages, graded against human labels), a globally corrected judge (Rogan–Gladen) is biased by several points and carries confidence intervals four to seven times wider than the multicalibrated estimator, and its bias grows as the language mix shifts; multicalibration stays within a point of the truth with tight intervals. Second, the correction works on the judge’s bare Yes/No labels, matching what it achieves on elicited $P(\text{helpful})$ scores: the gain comes from conditioning on features, not from eliciting confidences. Third, judge agreement is not the right target: our judge’s scores discriminate human labels only weakly (AUC 0.65), yet multicalibration still yields near-zero aggregate bias, because prevalence accuracy is governed by calibration rather than by how well the judge ranks responses. A controlled benchmark with exact ground truth (American Community Survey employment) and a simulation confirm the mechanism, and show it is classifier-agnostic.

Background and related work

LLM-as-judge and the reporting problem. Using one model to evaluate another is now routine, from MT-Bench and Chatbot-Arena-style preference judging (Zheng et al. 2023) to self-rewarding training loops (Yuan et al. 2024) and reward-model benchmarks (Lambert et al. 2024). The reliability of judges themselves is increasingly scrutinized, and a recent line of work targets the specific problem we study: reporting an unbiased *aggregate* from an imperfect judge. Lee et al. (2026) and Fiedler (2026) correct the raw judged rate with a Rogan–Gladen adjustment (Rogan and Gladen 1978) using judge sensitivity/specificity estimated on a labeled set, and construct confidence intervals; Lee et al. (2026) show unbiasedness under shift in the true-label distribution, i.e., $P(Y)$ may differ between calibration and test while $P(\hat{Y} | Y)$ is held fixed. Landesberg and Narayan (2026) calibrate judge scores to oracle labels via isotonic regression and audit transport per target. Li (2026) study calibrated aggregation across a panel of judges. These methods share the invariance we challenge: the judge’s conditional error behavior is treated as stable across populations, which is the appropriate model under label shift but not under the covariate shift that dominates deployment. We take these as the baselines to beat and show where and why they fail.

Quantification. Estimating a class proportion from imperfect predictions is *quantification* in machine learning (González et al. 2017) and the target of ReadMe (Hopkins and King 2010) and its successor (Jerzak et al. 2023). Classical quantifiers assume *label shift*, in which $P(X | Y)$ is stable while the prior $P(Y)$ may move: Adjusted Count, the Saerens–Latinne–Decaestecker (SLD/EMQ) algorithm (Saerens et al. 2002), and distribution-matching methods such as DyS and HDy (Maletzke et al. 2018). That is the right model when the label causes the features. When a judge scores a response, the features (model, task, language, content, length) drive the outcome, so $P(Y | X)$ is stable and $P(X)$ shifts. Under this covariate shift, label-shift quantifiers are not guaranteed unbiased, as our results confirm.

Using imperfect labels for valid inference. A complementary literature debiases a downstream estimate computed from cheap machine labels using a small gold sample drawn *from the target*: design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023). These are assumption-light (no covariate-shift or ignorability condition) but *target-*

specific by construction: each new population needs its own gold sample. That is the per-target cost that evaluation-at-scale, applying one judge to many models and tasks, is trying to avoid. We treat these as complementary and return to the trade-off in the discussion.

Calibration and its non-transfer. Calibration (predicted probabilities matching empirical frequencies) is the property that links a probabilistic scorer to an unbiased aggregate: a globally calibrated device has mean score equal to the population prevalence. But global calibration is a property of *one* population and need not survive a shift. LLM calibration degrades two- to three-fold under language shift even when accuracy is preserved (Yang et al. 2023) and varies widely across tasks (Ren et al. 2025). Confidence-elicitation methods (verbalized confidence (Tian et al. 2023), token log-probabilities (Kadavath et al. 2022), self-consistency (Wang et al. 2023)) and post-hoc recalibration (Oliveira et al. 2025) target *global* calibration and inherit this non-transfer. Multicalibration (Hébert-Johnson et al. 2018; Detommaso et al. 2024) conditions on features and is therefore stable under reweighting of those features; Detommaso et al. (2024) apply it to per-response LLM confidence scoring (hallucination detection), a per-instance objective distinct from the population-level aggregate we target. This is the property we exploit for aggregate evaluation.

Setup: aggregate estimation from an imperfect judge

Let $Y \in \{0, 1\}$ be the property of interest (e.g., a response is helpful, correct, or safe), X the observable features of the item being judged (model, task, language, length, content), and $h(X) \in [0, 1]$ the judge, which may emit a hard label in $\{0, 1\}$ or a score. The estimand is the aggregate $\pi^* = P^*(Y=1)$ in a target population (a pass rate, helpfulness rate, or violation rate), to be estimated from unlabeled target items $\{X_i\}$ and the judge, using a labeled *calibration* set drawn from a source population.

We assume *covariate shift* (Storkey 2009): $P(X)$ differs between calibration and target while $P(Y | X)$ is stable. This is the natural model when the features causally precede the property being judged (the model, task, and content of a response determine whether it is helpful, not vice versa). It is also the exact complement of the label-shift invariance that the Rogan–Gladden family assumes. Label shift and concept drift are discussed under limitations.

Why global corrections fail and multicalibration suffices

Global calibration is population-specific. A judge is *globally calibrated* on a population if $\mathbb{E}[Y | h(X)=p] = p$ for all p ; then $\mathbb{E}[h(X)] = \pi$ and averaging scores estimates the aggregate without bias. But calibration on one population need not hold on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the judge’s mean signed error in group G . Global calibration requires only that these errors cancel *on average*: $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the aggregate bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

nonzero in general unless every $\epsilon_G = 0$. A judge can be perfectly calibrated on the meta-evaluation set (its group errors offsetting under w_G) and badly biased on a new model or task that reweights those groups. High agreement offers no protection: AUC depends only on the ranking of scores and

is invariant to all miscalibration, including the feature-conditional errors ϵ_G that drive aggregate bias under shift.

Every global correction inherits this mechanism. Classify & Count (report the fraction the judge labels positive) and the Rogan–Gladen adjustment rely on error rates estimated on the calibration set; those rates are weighted averages over groups and change with target composition. The Rogan–Gladen estimate $\hat{\pi}_{\text{RG}} = (\hat{p} - \widehat{\text{FPR}})/(\widehat{\text{TPR}} - \widehat{\text{FPR}})$ is additionally *unstable*: when the judge is lenient, $\text{TPR} - \text{FPR}$ is small and its ratio form amplifies both bias and variance. SLD/EMQ (Saerens et al. 2002) and distribution-matching quantifiers assume label shift and fail under covariate shift. Global recalibration by isotonic regression removes average miscalibration but not the group-specific ϵ_G (Wu and Resnick 2024).

Feature-conditional calibration removes the bias. The fix is to drive every ϵ_G to zero, not merely their weighted sum. A predictor f is *multicalibrated* with respect to a group class $\mathcal{G}$ if $\mathbb{E}[Y \mid f(X)=v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018). The condition strictly needed for unbiased aggregates under every reweighting along $\mathcal{G}$ is the weaker *multi-accuracy*, $\mathbb{E}[f(X) - Y \mid X \in G] = 0$ for all $G \in \mathcal{G}$. Its consequence is the Kim et al. (2022) universal-adaptability guarantee: if f is multi-accurate with respect to $\mathcal{G}$, then for any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$,

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*.$$

One judge, calibrated once on source data, is thus unbiased on every target whose shift lies along the calibrated feature dimensions and within support: no knowledge of the shift, no target-specific estimation.

Why multicalibration rather than multi-accuracy. Multi-accuracy is the minimal sufficient condition, but multicalibration is the better practical target. Post-hoc algorithms produce it at no extra cost (we use MCGrad (Tax et al. 2026), which fits residual structure by gradient boosting in logit space), and it extends the guarantee to score-mediated shifts (e.g., selecting which items to human-audit by judge score, or reporting the aggregate within score strata) and provides robustness to misspecifying which features drive the shift (Gopalan et al. 2022).

The scope condition. The guarantee rests on two requirements. *Support/overlap*: the target’s feature values must lie within the calibration support. *Expressiveness*: the class $\mathcal{G}$ must be rich enough that within-cell $P(Y \mid X)$ is stable across populations: the features must capture every dimension along which the shift interacts with the outcome (an ignorability assumption). The support condition is the analogue of the positivity that importance weighting needs; the expressiveness condition is discharged once, at calibration time, rather than re-estimated per target. When either fails (a novel feature value, or latent variation no feature captures), estimates degrade, as our out-of-support results show.

Simulation

We first illustrate the mechanism in a setting simple enough to verify by hand. Data have a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y=1 \mid X=1) = 0.85$, $P(Y=1 \mid X=0) = 0.15$. A judge produces deterministic, systematically biased scores (a 10% multiplicative error within each stratum), wrong within each stratum but corrected to the true prevalence by a single global recalibration on the balanced training distribution $P(X=0) = 0.5$. We estimate the aggregate on targets with $P(X=0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed, over 50 replications.

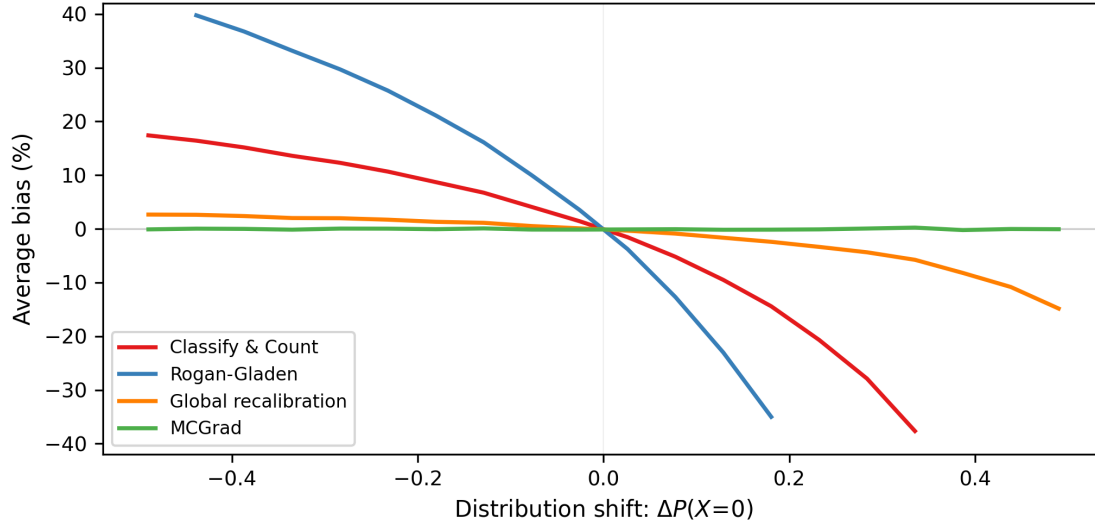


Figure 1. Relative aggregate bias under covariate shift (50 runs). The x-axis is the change in $P(X=0)$ from the training value of 0.5. Classify & Count and Rogan–Gladen diverge with shift; global recalibration shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; full range and additional methods in the appendix.

At the training distribution all methods are approximately unbiased (Figure 1). As the target shifts they diverge: Rogan–Gladen is most unstable, its ratio form amplifying error beyond $\pm 40\%$; Classify & Count grows in the same direction; global recalibration shows moderate but nonzero bias; the multicalibrated estimator maintains near-zero bias and the lowest RMSE across the entire range.

LLM-as-judge helpfulness under language shift

Our main study puts the argument in the setting that motivates it: an LLM judge deployed across a shifting population of responses, with human labels as ground truth. We use the OpenAssistant corpus (Köpf et al. 2023), which provides assistant responses annotated by multiple human reviewers for helpfulness. The estimand is the *helpfulness rate*, the fraction of responses in a target population that humans consider helpful. We binarize the human helpfulness score (mean of ≥ 3 reviewers) at 0.6 as the gold label, and take Claude Sonnet as the judge. The shift dimension is *language*: a judge calibrated on one language mix is applied to a different one, the direct analogue of a judge calibrated on today’s models being applied to a new model, and the setting where LLM calibration is known to degrade sharply (Yang et al. 2023).

Design. We sample 17,211 assistant responses across six languages (English, Spanish, Russian, German, French, Chinese). The human-gold helpfulness rate varies strongly by language, from 0.32 (Chinese) to 0.71 (German), so reweighting the language mix moves the target aggregate, which is the covariate-shift regime. We compare two judge output modes spanning current practice: a discrete Yes/No helpfulness label (the standard in applied use) and a directly elicited $P(\text{helpful})$ score. We calibrate on four languages (English, Spanish, Russian, German) and hold out two (French, Chinese) as out-of-support targets that share nothing with the calibration languages, a deliberately hard test. MCGrad conditions on observable metadata only: language, response length (log characters), and the judge’s own label or score. We benchmark against Classify & Count, Rogan–Gladen (the current

judge-reporting recommendation, with a single global sensitivity/specificity), importance-weighting (IPW), isotonic regression, and SLD.

The judge is a realistic, imperfect instrument. It is systematically lenient (it labels 87% of responses helpful against a 62% gold rate), and its error rates vary by language: per-language sensitivity ranges 0.83–0.98 and specificity 0.07–0.36. Because error rates differ by language, a single global Rogan–Gladen correction *cannot* remain valid once the language mix moves; this is the premise of the experiment, confirmed in the data. The judge’s elicited scores discriminate the human gold only weakly (AUC 0.65): helpfulness is subjective, and the judge agrees with human reviewers far less than a topic classifier would. This makes the study a stress test of the central claim that *prevalence accuracy is a property of calibration, not discrimination*.

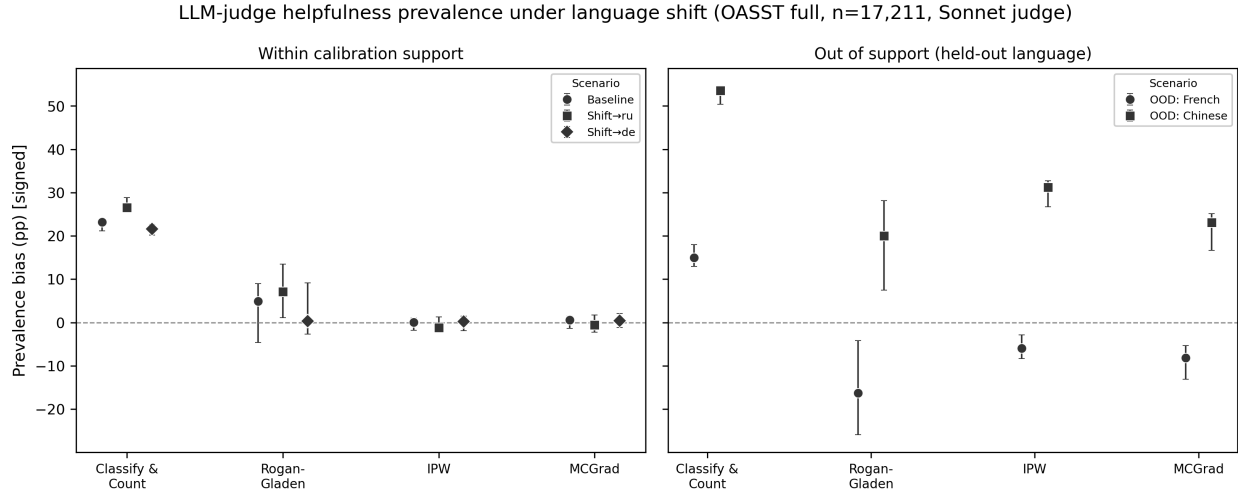


Figure 2. Aggregate helpfulness bias (signed percentage points, with 95% bootstrap confidence intervals) for a Sonnet judge on 17,211 OpenAssistant responses. Left: within calibration support (baseline plus two language-mix shifts). Right: out of support (an entirely held-out language). Markers denote scenarios; bootstrap resamples the calibration set and refits. Classify & Count is reliably biased; Rogan–Gladen is biased and high-variance; IPW and MCGrad are near-zero within support, with MCGrad requiring no per-target estimation.

Results (Figure 2, full scale). Within support, the pattern is clear. Classify & Count (the default of simply counting the judge’s positive labels) is biased by +22 to +27pp across scenarios. Rogan–Gladen carries a 4pp mean absolute bias whose confidence intervals are four to seven times wider than MCGrad’s, and its bias *grows* as the mix shifts toward the language where the judge is most lenient (Russian: +4.9 → +7.1pp). Its instability is intrinsic: the lenient judge gives $\text{TPR} - \text{FPR} \approx 0.13$, so the ratio correction amplifies calibration noise. IPW and MCGrad both achieve near-zero bias (mean $|\cdot| \leq 0.5\text{pp}$) with tight intervals ($\pm \sim 1.5\text{pp}$). Out of support (a held-out language absent from calibration), every method degrades, as the scope condition predicts: Rogan–Gladen swings wildly (−16pp on French, +20pp on Chinese), and MCGrad is the most stable on average (OOD mean $|\cdot|$ 15.7pp vs. IPW 18.6, Rogan–Gladen 18.2, Classify & Count 34.3) but is not immune: Chinese, with a 0.32 gold rate and very short responses, is hard for all methods.

MCGrad versus IPW. Within support, MCGrad and IPW tie on bias, as the theory predicts, since both rest on the same ignorability assumption and both are unbiased when the calibration features capture the shift. The contrast is not accuracy but *cost and reusability*. IPW re-estimates density ratios *separately for every target*; the near-zero IPW numbers were bought by fitting a

new source-vs-target model for each scenario. MCGrad is fit once and applied to all targets, the universal-adaptability property (Kim et al. 2022). As the number of evaluated models, tasks, or languages grows, IPW’s cost scales linearly while MCGrad’s is constant; the tie on a five-target table is the *confirmation* that MCGrad matches per-target-optimal IPW without per-target work. IPW additionally fails under positivity violations, which we exhibit sharply in the controlled benchmark below.

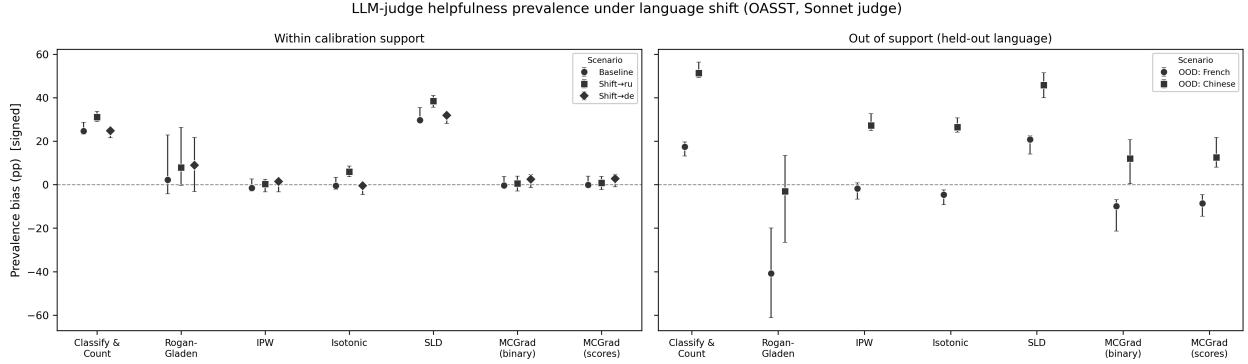


Figure 3. The full method comparison at pilot scale (6,000 responses, both judge output modes, 95% bootstrap CIs). Within support, MCGrad on bare Yes/No labels (MC binary) and on elicited $P(\text{helpful})$ (MC scores) are indistinguishable ($\leq 1.2\text{pp}$), both near zero; SLD (a label-shift quantifier) fails by +30 to +38pp; isotonic (global recalibration) is decent within support but degrades out of it.

Binary matches scores; label-shift quantifiers fail (Figure 3). At pilot scale we run the full method suite including both judge output modes. MCGrad on the judge’s bare Yes/No labels achieves the same within-support bias as MCGrad on elicited $P(\text{helpful})$ scores (both $\leq 1.2\text{pp}$; OOD 10.9 vs. 10.6pp): the correction comes from conditioning on features and the calibration design, not from eliciting confidences, so practitioners can apply it to the Yes/No labels they already collect. SLD, the canonical label-shift quantifier, is badly biased throughout (+30 to +38pp), confirming that the label-shift family is the wrong invariance for judge evaluation. Isotonic regression (global recalibration) is reasonable within support (2.3pp) but degrades out of it (15.5pp), because global calibration does not transfer. The headline within-support ordering is robust to the gold binarization threshold: repeating the analysis at a threshold of 0.75 (a balanced 50% base rate) leaves MCGrad and IPW near zero ($\leq 1.0\text{pp}$) while Rogan–Gladen worsens to 18.8pp, as the judge’s leniency mismatch grows.

Controlled validation with known ground truth

The OpenAssistant study is the realistic demonstration; this benchmark is the identification-clean confirmation, where ground truth is known exactly and the base classifier is a conventional model rather than an LLM (establishing that the mechanism is classifier-agnostic). Using the American Community Survey via *folktables*, we estimate the *employment rate* from 16 sociodemographic features with logistic regression, training on eight states (2016–2018, $\approx 1.5\text{M}$ observations), calibrating on a held-out set ($n \approx 644,000$), and evaluating both in-distribution and on six held-out states. Because employment varies sharply by age (76% for ages 25–54 vs. 17% for 65+), we construct covariate shifts by resampling the target to be young-skewed, old-skewed, or bimodal, yielding true rates from 12.8% to 46.0%; all calibration parameters are fixed across scenarios.

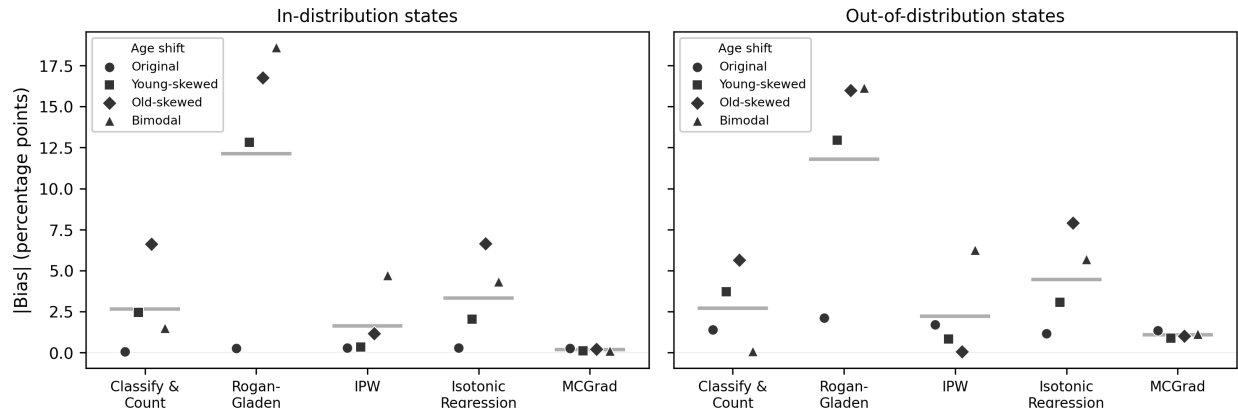


Figure 4. Aggregate employment-rate bias (absolute pp) under age-distribution shift, in-distribution states (left) and out-of-distribution states (right). Markers denote shift scenarios.

The pattern matches the simulation and judge study (Figure 4). Rogan–Gladen fails by 12–19pp and SLD comparably; Classify & Count and isotonic show moderate but growing bias (up to 8pp). MCGrad achieves ≤ 0.27 pp bias across all in-distribution scenarios, including the bimodal shift, and degrades only modestly out of distribution (0.88–1.35pp), reflecting geographic shift along a dimension the calibration set did not span. This is where the IPW contrast becomes an empirical win, not just a workflow one: IPW is excellent on simple shifts (≤ 1.2 pp) but fails on the *bimodal* shift (+4.7pp in-distribution, +6.3pp out) where the density ratio is hard to model, a positivity stress that MCGrad absorbs. The consistent story across an exact-oracle benchmark and a noisy-human-label judge study is that feature-conditional calibration is near-unbiased when the target lies within calibration support and degrades predictably when it does not.

Recommendations for LLM-judge evaluation

The result yields a concrete protocol that fits how evaluations are already run.

1. **Judge as usual.** Prompt the judge for a per-response label (or score). No confidence elicitation or model internals are required; bare Yes/No suffices.
2. **Collect a calibration set that spans the shift dimensions.** Human-label a modest sample that *covers* the dimensions along which your targets will differ: model family, task domain, language, response length. Coverage of these dimensions, not raw size, is what the guarantee needs.
3. **Multicalibrate on observable metadata** (plus the judge’s label/score), using any multi-accuracy/multicalibration fitter. This needs no access to the judged model’s weights or logits.
4. **Report the aggregate by averaging calibrated scores** on any target within the calibration support, with no refitting. Bootstrap the full pipeline (resample the calibration set, refit, recompute) for confidence intervals.

Which tool when. If you can draw a fresh gold sample *from each target* you evaluate, design-based methods and prediction-powered inference (Egami et al. 2023; Angelopoulos et al. 2023) are efficient and need no covariate-shift assumption; prefer them. If instead you evaluate *many* targets (the

leaderboard and continuous-evaluation situation), multicalibration provides one reusable instrument at the cost of the ignorability/support assumption. Rogan–Gladden and SLD are appropriate under label shift, but LLM-judge deployment is a covariate-shift regime, where they are biased and, for lenient judges, unstable.

Diagnostics. Because the guarantee is support-limited, verify coverage rather than reporting agreement alone: cross-tabulate the target’s feature cells (model $\times$ domain $\times$ language, plus binned length) against the calibration cells and flag any target cell empty in calibration; those are extrapolation. Report AUC if you like, but do not treat it as evidence the aggregate transports; a judge with AUC 0.65 can still yield a near-unbiased aggregate under multicalibration, and a judge with AUC 0.99 can be badly biased under shift.

Discussion

Aggregates produced by LLM judges (pass rates, win rates, safety-violation rates) increasingly decide which models ship and which are deemed safe, yet they are more fragile than current reporting acknowledges. Agreement metrics certify that a judge separates classes; they are silent on whether the *number* transports to the model or task being evaluated. The recently recommended fix, Rogan–Gladden bias correction, transports only under label shift; deployment is a covariate-shift regime, where it is biased and, for the lenient judges we observe, high-variance. We have shown that calibration conditional on features (multi-accuracy as the minimal condition, multicalibration as its practical realization) is what makes the aggregate transport, turning a validated judge into an instrument calibrated once and applied to many populations.

The central practical payoff is target-independence. Importance weighting, design-based methods, and prediction-powered inference all redo work per target; a multicalibrated judge is calibrated once and deployed without target-specific estimation, which is the property continuous, many-model evaluation most needs. That advantage is bought with an assumption: the calibration features must capture the shift and targets must lie in support. We view stating this scope condition, and showing its bite through honest out-of-support results, as a strength.

Several limitations follow. First, the guarantee holds only along calibrated dimensions; a novel feature value (a language, a model family) absent from calibration degrades all methods, so calibration data must be designed to span anticipated variation. Our hardest out-of-support case, an entirely held-out language, degrades every estimator, and reporting this honestly is the point. Second, multicalibration needs labeled calibration data; when per-target gold labels are cheap, design-based methods are the better tool and the two compose. Third, we assume covariate shift; under concept drift ($P(Y | X)$ itself changing, for instance human standards for “helpful” evolving) no purely statistical correction substitutes for new labels. Finally, our judge study uses human helpfulness labels as ground truth, which are themselves noisy; we report bootstrap intervals throughout and confirm the mechanism on an exact-oracle benchmark. The agreement metrics the field reports remain useful, but for a trustworthy aggregate across the distributions that motivate LLM-based evaluation, they are not sufficient.

Reproducibility and experimental details

Simulation. $n = 10,000$ per run, $P(Y=1 \mid X=1) = 0.85$, $P(Y=1 \mid X=0) = 0.15$; judge scores $\hat{p}(X=0) = 0.135$, $\hat{p}(X=1) = 0.935$; 50 runs; targets $P(X=0) \in [0.01, 0.99]$. Full method definitions in the appendix.

LLM-as-judge (OpenAssistant). 17,211 assistant messages with human helpfulness labels across six languages (English, Spanish, Russian, German, French, Chinese; English/Spanish capped at 4,000, the others taken in full). Gold label: human helpfulness ≥ 0.6 (robustness at 0.75 reported). Judge: Claude Sonnet, in two independent campaigns: binary Yes/No and direct $P(\text{helpful})$ elicitation without prior commitment to an answer. Each response is scored with the codebook helpfulness definition and its preceding conversational context; the judge never sees the human gold label. MCGrad conditions on language, log response length, and the judge’s label (base-rate initialization) or score. Calibration draws from English/Spanish/Russian/German; French and Chinese are held out as out-of-support targets. Baselines: Classify & Count, Rogan–Gladen (global sensitivity/specificity), IPW (density ratios via logistic regression on language and log length), isotonic regression, and SLD. Confidence intervals via 200-iteration full-pipeline bootstrap (resample calibration set, refit calibrator and error rates, recompute). The full-scale comparison (Figure 2) uses the binary campaign; the complete method suite including elicited scores (Figure 3) uses a 6,000-response subset run in both modes.

American Community Survey. *folktables* ACSEmployment task, 16 features; logistic regression with standard scaling; train on TX, MI, PA, OH, IL, GA, NC, VA (2016–2018); calibrate on a held-out set ($n \approx 644,000$); test in-distribution and on CA, NY, FL, WA, AZ, CO. Age-shifted targets via importance-weighted resampling; RMSE via 200-iteration bootstrap. Post-hoc calibration by isotonic regression and MCGrad.

Multicalibration algorithm. Claims are algorithm-agnostic: any procedure producing a multi-accurate (a fortiori multicalibrated) predictor over the feature class delivers the guarantee (Hébert-Johnson et al. 2018; Gopalan et al. 2022; Detommaso et al. 2024). We use MCGrad (Tax et al. 2026), which fits gradient-boosted trees on label residuals in logit space with the current prediction as an input feature, so splits discover miscalibrated regions in the joint space of features and score levels. An open-source implementation is at <https://mcgrad.dev>.

Data and code. Replication code is available at https://github.com/facebookresearch/multicalibrated_llm_measurement. ACS data is public via *folktables*; the OpenAssistant corpus is public (Köpf et al. 2023). Judge inputs and outputs for all 17,211 responses (both campaigns) are included in the replication materials, along with the judge prompt templates.

Competing interests. Several authors are affiliated with Meta Platforms, and a subset are authors of the MCGrad algorithm used here; the open-source implementation is maintained by their team. The conceptual claim and the empirical guarantee are algorithm-agnostic, and any multi-accuracy or multicalibration fitter can be substituted for MCGrad.

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